Problem Set 1

Grant Talbert
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Boston University

Problem 1

Problem 2

Let $x^\mu = (x, y, z)$ be the cartesian coordinate system in $\mathbb{R}^3$, and (r, θ, ϕ) the spherical coordinate system

$$x = r \sin\theta \cos\phi, \qquad y = r \sin\theta \sin\phi, \qquad z = r\cos\theta.$$

The metric then takes the form

$$\mathrm{d}s^2 = \mathrm{d}r^2 + r^2 \mathrm{d}\theta^2 + r^2 \sin^2\theta \mathrm{d}\phi^2.$$

Problem 2.1. *A particle moves along a curve*

$$x^\mu(\lambda) = (x(\lambda), y(\lambda), z(\lambda)) = (\cos\lambda, \sin\lambda, \lambda).$$

Express the path of the particle in the (r, θ, ϕ) coordinate system.

Solution. This feels too easy and like a trick question. From simple geometric reasoning, we would think that

$$r^2 = x^2 + y^2 + z^2,$$

$$\phi = \arctan\frac{y}{x}, \qquad \theta = \arctan\frac{\sqrt{x^2 + y^2}}{z^2}.$$

Some care must be taken to ensure the arctangent is taken in the proper quadrant.